**Q NO: 01**

**A computer graphics application needs to manage circles at different positions on a coordinate plane. Each circle has a radius and a center position (x, y). Design and implement a Circle class in C++ to solve this problem. Your solution should:**

**• Include three attributes: radius, x, and y.**

**• Create a no-argument constructor to initialize all attributes to 0. • Create a three-argument constructor to initialize x, y, and radius with user-provided values.**

**• Create a void setValues(float, float, float) function to update the circle's values.**
**• Create float area() and float circumference() functions to calculate and return the area and circumference.**

**• Create a void print() function to display the circle's x, y, and radius.**

**• In main(), create appropriate Circle objects and call the functions to test their working.**

**• While testing, consider: What should happen if the user provides an invalid value, such as a negative radius?**

**How your design demonstrates technical competence, responsible programming, and thoughtful decisionmaking in solving the given problem.**

```
#include <iostream>

using namespace std;


class Circle

{

private:

   float radius;

   float x;
```

```
    float y;

public:
    Circle()
    {
        radius = 0;
        x = 0;
        y = 0;
    }

    Circle(float xValue, float yValue, float r)
    {
        x = xValue;
        y = yValue;

        if (r >= 0)
            radius = r;
        else
        {
            cout << "Invalid radius!" << endl;
            radius = 0;
        }
    }

    void setValues(float xValue, float yValue, float r)
    {
```

```
    x = xValue;
    y = yValue;

    if (r >= 0)
        radius = r;
    else
    {
        cout << "Invalid radius!" << endl;
        radius = 0;
    }
}

float area()
{
    return 3.14159 * radius * radius;
}

float circumference()
{
    return 2 * 3.14159 * radius;
}

void print()
{
    cout << "X = " << x << endl;
    cout << "Y = " << y << endl;
```

```
        cout << "Radius = " << radius << endl;
        cout << "Area = " << area() << endl;
        cout << "Circumference = " << circumference() << endl;
    }
};

int main()
{
    Circle c1;
    Circle c2(5, 10, 7);

    cout << "Circle 1:" << endl;
    c1.print();

    cout << "\nCircle 2:" << endl;
    c2.print();

    cout << "\nAfter setting values:" << endl;
    c1.setValues(2, 3, 4);
    c1.print();

    return 0;
}
```